

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama Belgaum-590018



A Report On "PHYSICAL EDUCATION"

Submitted in partial fulfillment of the requirements for 6th semester of

Bachelor of Engineering in
Information Science & Engineering

Submitted by:

ROHITH G (1AM23IS132)

Under the Guidance of Mr.Munikrishna G T

Physical Education Director

Dept of Sports, AMCEC



DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

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AMC ENGINEERING COLLEGE

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BELGAUM) BENGALURU-560083

Information Science and Engineering



CERTIFICATE

This is to certify that **ROHITH G (1AM23IS132)**, the Bonafide student of AMC Engineering College has successfully carried out her Physical Education Activities in partial fulfillment of the requirements for the award of degree in Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University, Belgaum during academic year 2025-2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The AICTE report has been approved as it satisfies the academic requirements in respect of the said degree.

Dr. Rashmi R Deshpande

Professor and Head

Department of ISE

Mr Munikrishna G T

PED

Dept of Sports

Dr Yuvraju B
N

Principal

AMCEC

DECLARATION

I, ROHITH G student of 6th semester Bachelor of Engineering at the Department of Information Science and Engineering, AMC Engineering College, Bengaluru declare that the dissertation work entitled "Physical Education" (Physical Education) has been carried out by me and submitted in partial fulfilment of the course requirement for the award of degree in Bachelor of Engineering in Information Science & Engineering of Visvesvaraya Technological University, Belagavi during the year 2025-26. I also declare that to the best of my knowledge and belief, the work reported here does not form any part of other dissertation based on which a degree or award was conferred in earlier occasion on this by any other student.

ROHITH G
(1AM24IS132)

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(1AM23IS132)

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1	Module 1	20	
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Physical Education Director

INTRODUCTION

Physical education is an integral component of a holistic education system, emphasizing the importance of physical fitness, mental well-being, and social interaction. It serves as a foundation for promoting an active lifestyle, fostering teamwork, and enhancing the overall quality of life. The primary objective of physical education is not just to engage in physical activities but also to understand the principles of health, fitness, and sportsmanship.

The evolution of physical education over the years has seen the integration of structured exercises, diverse sports, and innovative training methods to cater to individual and group needs. This discipline plays a critical role in shaping the physical, cognitive, and emotional development of individuals, enabling them to lead healthier and more fulfilling lives.

In the context of academics, physical education offers students a break from sedentary activities, encouraging movement and recreation. Through activities like team sports, athletics, and fitness regimens, it instills essential life skills such as discipline, resilience, and cooperation. Moreover, it bridges the gap between theory and practice, providing a hands-on approach to learning about nutrition, exercise physiology, and sports techniques.

This report delves into various aspects of physical education, as outlined in the curriculum, including foundational fitness principles, the importance of nutrition, specific fitness assessments, and the mastery of selected sports or athletic events. Each chapter aims to provide a comprehensive understanding of the topics while highlighting their practical significance. Physical education is not just about excelling in sports; it is about nurturing a well-rounded individual who values health, teamwork, and lifelong physical activity. This report is a step toward recognizing and embracing the benefits of this essential discipline.

Chapter-1

1.1 Postural deformities

1.1.1 Meaning & Importance

Postural deformities refer to abnormal alignment or curvature of the spine and body posture that can negatively affect a person's overall physical health and appearance. Common types of postural deformities include kyphosis (a hunched upper back), lordosis (an exaggerated inward curve of the lower back), and scoliosis (a sideways curvature of the spine). These conditions may arise due to poor posture habits, weak muscles, injuries, congenital issues, or improper physical activities. Understanding postural deformities is important in physical education because early detection and correction can prevent long-term health problems such as chronic pain, limited mobility, and reduced physical performance. It also plays a crucial role in improving students' awareness of their body alignment, which helps enhance balance, coordination, and efficiency in movement. Good posture supports better breathing, reduces fatigue, and increases confidence, all of which contribute to overall physical and mental well-being. Physical education programs aim to educate students about maintaining correct posture through proper exercises, body mechanics, and lifestyle adjustments, thereby promoting lifelong musculoskeletal health.

1.1.2 Types of Postural Deformities

There are three major types of postural deformities. Kyphosis refers to an excessive outward curve of the upper spine, resulting in a rounded or hunched back. Lordosis is the exaggerated inward curve of the lower back, often seen in individuals with weak abdominal muscles. Scoliosis is a condition where the spine curves sideways in an “S” or “C” shape, leading to uneven shoulders or hips. Each of these deformities affects the body's balance, flexibility, and appearance and may lead to discomfort or pain over time if not addressed.

1.1.3 Causes of Postural Deformities

Postural deformities can be caused by a variety of factors. Poor sitting and standing habits, carrying heavy backpacks incorrectly, or maintaining a sedentary lifestyle can lead to changes in body posture. Additionally, lack of exercise, improper ergonomics, weak muscles, or tight muscle groups can contribute to the development of these deformities. In some cases, they are present from birth (congenital) or may result from injuries or diseases like polio and rickets.

1.1.1 Prevention and Correction

Postural deformities can often be prevented and corrected through regular exercise, stretching, and strengthening of core muscles. Activities like yoga, Pilates, and physiotherapy are beneficial in maintaining proper posture. Teaching students about the importance of good posture, correct sitting and standing techniques, and regular movement can help reduce the risk of developing deformities. In more severe cases, medical treatment, braces, or surgery may be needed, but most mild to moderate cases can be improved through physical activity and awareness.

1.2 Stress Management

1.2.1 Introduction to Stress and Stress Management

Stress is a natural physical and emotional response that the body exhibits when faced with challenges, demands, or pressures. It is a part of everyday life and can arise from both positive and negative situations. While a certain level of stress is normal and can even motivate individuals to perform better, excessive or unmanaged stress can become harmful. In today's fast-paced world, especially for students, stress has become a common concern. Stress management refers to the various methods, techniques, and life strategies used to handle stress effectively, ensuring that it does not adversely impact one's mental, emotional, or physical health.

1.2.2 Causes of Stress Among Students

Students face numerous sources of stress in their academic and personal lives. Academic pressure to perform well in exams, maintain high grades, and meet deadlines can be overwhelming. Apart from studies, expectations from parents and teachers, competition with peers, and fear of failure can also contribute to rising stress levels. Social factors such as peer pressure, friendship issues, bullying, or social media influence can deeply affect students' emotional well-being. In addition, poor time management, lack of physical activity, unhealthy diet, and sleep deprivation further aggravate stress. Adolescents also go through physical and hormonal changes that can lead to mood swings and emotional instability.

1.2.3 Effects of Stress on Health and Academic Performance

Chronic or unmanaged stress can have serious consequences on a student's overall health. Physically, it may result in fatigue, frequent headaches, back pain, sleep disturbances, appetite changes, and even long-term health conditions like high blood pressure. Psychologically, stress can cause anxiety, irritability, low self-esteem, difficulty concentrating, and lack of motivation. These symptoms can negatively affect academic performance, making it harder for students to focus, retain information, and perform to their full potential. Long-term stress can also lead to emotional burnout and mental health disorders such as depression or panic attacks if not managed appropriately.

1.2.4 Importance of Stress Management in Physical Education

Physical education plays a vital role in promoting stress management among students. Regular participation in physical activities such as sports, games, running, yoga, and aerobic exercises helps reduce stress hormones like cortisol while increasing the production of endorphins—the body's natural mood lifters. Engaging in physical activity improves blood circulation, boosts energy levels, and promotes better sleep, all of which are essential in managing stress. Furthermore, physical education encourages teamwork, discipline manner.

1.2.5 Effective Stress Management Techniques

There are several proven techniques to manage stress effectively. Physical exercises like walking, swimming, dancing, and cycling not only keep the body fit but also help release tension.

Breathing exercises, meditation, and yoga help calm the mind and promote relaxation. Maintaining a well-organized routine, setting realistic academic goals, and avoiding procrastination reduce the feeling of being overwhelmed. Healthy eating, drinking enough water, and getting sufficient sleep are basic but essential habits to keep stress at bay. Students should also take time for hobbies, socialize with supportive friends, and practice mindfulness. Positive self-talk, avoiding negative thinking, and seeking help from teachers, counselors, or psychologists when needed are also crucial parts of managing stress successfully.



Fig 1.1

Chapter - 2

2.1 Table Tennis

2.1.1 Fundamental Skills

Table Tennis, also known as ping pong, is a fast-paced indoor sport that requires quick reflexes, agility, and excellent hand-eye coordination. It is played on a rectangular table divided by a net, and the objective is to hit the lightweight ball back and forth using small rackets (also called paddles), aiming to outscore the opponent through precise shots and strategic placement. To succeed in this sport, mastering a range of fundamental skills is essential.

Another important basic is stance and footwork. A correct stance keeps the player balanced and ready to move in any direction. The feet should be shoulder-width apart, knees slightly bent, and the body leaning slightly forward. Good footwork allows players to position themselves correctly before each shot. Quick side – to – side and forward - backward movements are essential to reach the ball efficiently and maintain rhythm during fast- paced rallies.

Topspin strokes, including the forehand and backhand topspin, are advanced offensive techniques that allow players to attack with speed and spin. By brushing up on the ball at a sharp angle, players can create topspin that causes the ball to dip and bounce high off the table, making it difficult to return. Mastering topspin is key to offensive play, especially at higher levels of competition.

Lastly, developing good rallying skills and game awareness is crucial. Rallying helps players maintain consistency under pressure, while understanding opponent patterns, spin types, and match tactics improves shot selection. Mental toughness, patience, and the ability to switch between offensive and defensive play styles are what separate average players from skilled competitors.

2.1.2 Game practice with application of Rules and Regulations

Game practice in table tennis is essential for applying the skills learned during training sessions in a real match setting. It not only helps players enhance their performance but also builds strategic understanding, mental focus, and the ability to make quick decisions.

While engaging in game practice, it is crucial to follow the official rules and regulations set by recognized bodies like the International Table Tennis Federation (ITTF). A standard match is played either to 11 points (best of 5 or 7 games), and a player must win by at least a 2-point margin. The serve alternates every two points between players, and in case of a 10- 10 tie (deuce), players alternate serves after each point. A legal serve involves tossing the ball straight up at least 16 cm and striking it so that it bounces once on the server's side and then on the receiver's side. In conclusion, game practice with application of rules and regulations bridges the gap between training and actual competition. It instills discipline, sharpens technical and tactical execution, and prepares players to perform confidently within the framework of official rules, ensuring a fair and competitive game environment.

Chapter – 3

3.1 Aerobics



Fig 3.1

3.1.1 Introduction to Aerobics

Aerobics is a system of physical exercise that combines rhythmic aerobic movement with stretching and strength training routines, typically performed to lively music. The term "aerobic" means "with oxygen," referring to exercises that stimulate heart and lung activity for a sustained period. Aerobic exercises are designed to improve the efficiency of the cardiovascular system in absorbing and transporting oxygen. These workouts are structured to enhance overall fitness, including cardiovascular endurance, muscular strength, flexibility, and coordination. Because they involve continuous movement, aerobic exercises are effective for improving stamina and burning calories. Aerobics is suitable for people of all ages and fitness levels and can be done in a group class, at home, or in a gym setting.

3.1.2 Types of Aerobic Exercises

Aerobic exercises can vary in intensity and style, and they cater to a wide range of fitness preferences and goals. Basic types of aerobic activities include walking, jogging, cycling, swimming, stair climbing, and dancing. In structured aerobic classes, participants perform choreographed steps, combining arm and leg movements to the beat of energetic music. These sessions may include warm-up, workout, and cool-down phases. Low-impact aerobics involve

gentle movements with at least one foot always on the ground and are ideal for beginners or elderly individuals. High-impact aerobics includes more vigorous activities like jumping, hopping, and running in place, which help in burning more calories and increasing strength. Other forms such as water aerobics, step aerobics, Zumba, and cardio kickboxing offer variety and keep participants engaged and motivated.

3.1.3 Physical and Mental Benefits of Aerobics

Aerobics provides a wide range of health benefits for both the body and the mind. One of the key advantages is improved cardiovascular health. Regular aerobic exercise strengthens the heart muscles, improves lung function, and enhances blood circulation throughout the body. It helps regulate blood pressure, reduces bad cholesterol levels, and increases good cholesterol levels. Aerobic activity also supports weight management by burning excess calories and body fat. Moreover, it increases flexibility, tones muscles, improves joint movement, and boosts physical stamina. On the mental health front, aerobics is known to be a natural mood booster. It reduces stress, anxiety, and depression by increasing the production of endorphins—the body's "feelgood" hormones. People who practice aerobics regularly often report better sleep, increased energy levels, sharper focus, and higher self-esteem.

3.1.4 Role of Aerobics in Physical Education

Aerobics holds a significant place in physical education programs due to its accessibility, health benefits, and fun nature. It provides students with a structured yet enjoyable way to stay active. Aerobics classes in schools and colleges are often designed to be energetic and engaging, incorporating popular music and group participation, which increases motivation and social interaction. These classes help students build discipline, learn coordination, and understand the value of physical fitness in everyday life. Incorporating aerobics into physical education encourages students to adopt a healthy and active lifestyle from a young age, helping them to prevent lifestyle-related health problems such as obesity, diabetes, and heart disease.

Furthermore, aerobic training teaches the importance of warm-up and cool-down routines, hydration, and safe movement techniques.

CONCLUSION

In conclusion, Physical education is not just about physical fitness but also about nurturing awareness, discipline, and preventive health care among students. One important aspect covered in physical education is postural deformities, which are irregularities in the alignment of the spine or body posture. Conditions like kyphosis, lordosis, and scoliosis can develop due to poor habits, lack of physical activity, or structural issues. Identifying these deformities early and taking corrective actions through proper exercises, postural awareness, and medical guidance helps prevent long-term complications and enhances the quality of life.

Another major component of student well-being is stress management, which is becoming increasingly relevant in today's fast-paced academic environment. Physical education teaches stress-relief techniques such as deep breathing, time management, meditation, and especially physical exercise—all of which help reduce anxiety and promote a calm, focused mind. When students are taught to handle stress constructively, it not only improves their mental health but also their academic performance and personal relationships.

Aerobics, as a part of physical education, brings both fun and fitness into a student's routine. It improves cardiovascular health, enhances stamina, and helps maintain a healthy body weight. Moreover, it boosts self-esteem, coordination, and energy levels while also reducing stress. Together, these topics demonstrate that physical education is essential for holistic development, helping students build strong bodies, balanced minds, and healthy lifestyles that benefit them both now and in the future. The study of these various topics has equipped students with the knowledge necessary to improve both their physical fitness and athletic skills. In future, as students advance in physical education, they will continue to build on this foundation, honing their abilities and pursuing excellence in various sporting disciplines.

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